

Forecasting Systemic Tension: A Machine Learning Approach to Spain's Grid Risk Index

Machine Learning Operations Project Proposal

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1. Executive Summary

The Spanish electrical grid operates under strict physical constraints, requiring constant, delicate balance between energy generation and demand to maintain frequency stability. Grid operators currently face a critical gap: the absence of strategic, day-ahead risk estimation. While static forecasts for demand and generation exist, they fail to quantify the overall *systemic tension* the grid will face the following day. This exposes infrastructure to sudden shocks, particularly when inflexible generation cannot ramp up quickly enough to meet unexpected demand spikes or drops in renewable output.

As the penetration of intermittent renewable energy sources deepens, the grid becomes increasingly volatile. Relying on isolated metrics, such as raw demand, without synthesizing them into a unified measure of systemic tension leaves the network vulnerable to costly, last-minute interventions or partial blackouts. The primary stakeholder is **Red Eléctrica de España (REE)**, the national transmission system operator, alongside energy policymakers at the Ministry for Ecological Transition and the CNMC. REE operations staff use the next-day risk score to plan flexible-source dispatch; policymakers use the weekly digest to identify recurring stress patterns and their infrastructure implications.

We propose a predictive machine-learning framework that forecasts a **Next-Day “Grid Risk Index.”** This index synthesizes the expected-versus-actual mismatch in demand and generation alongside the scheduled flexibility ratio of the energy mix. By providing a clear, day-ahead forecast using daily-aggregated historical data, the system enables REE operators and policymakers to transition from reactive troubleshooting to proactive resource planning. The early-warning mechanism ensures uninterrupted energy supply, optimizes strategic-reserve dispatch, and reduces overall operational costs.

2. Business Case & Objectives

2.1 Business Case

The primary value of this project is **proactive risk management** for national energy infrastructure. As the electrical grid approaches maximum capacity, operators must rely on costly measures such as activating backup gas plants or reducing the power to industrial users. A preventive risk-forecasting system provides an early-warning mechanism, allowing REE to plan flexible-energy-source dispatch more efficiently, reducing operational costs and improving network-wide stability.

2.2 Project Objectives

The project aims to build a **Gradient Boosted Tree** model (a continuous regressor using the LightGBM framework) that predicts a Next-Day Grid Risk Index ranging from 0 (stable) to 1 (high systemic operational stress). The core objectives are the following:

- **Minimize forecast error** (RMSE) so that predictions remain reliable for decision-makers.
- **Prioritize interpretability** by exposing feature-importance metrics so operators understand which variables drive forecast risk.
- **Simulate daily batch processing** using fixed historical datasets, removing infrastructure complexity and keeping focus on model quality.
- **Meet latency requirements:** The full inference pipeline (feature computation, index construction, and GBT prediction) must complete within five minutes to support REE’s day-ahead scheduling window, which closes at 22:00 CET.
- **Enable proactive management** transition from reactive troubleshooting to data-driven grid management, minimizing network impact, and avoiding emergency interventions.

Explicit success criteria: The primary technical success criterion is achieving an RMSE below 0.05 on the normalized 0–1 scale on the held-out chronological test set. This threshold will be confirmed

after the baseline run in Days 4–6 and locked before GBT training begins. For operational interpretation, the next-day predicted risk scores are mapped into *Low*, *Medium*, and *High* categories using validation-calibrated thresholds. The alerting quality of the *High* category is assessed on data using recall and precision; recall is prioritized because failing to flag a genuinely *High*-risk period has the greatest potential impact on grid stability.

These ML objectives directly support the business case: a model meeting the RMSE target reduces the frequency of costly last-minute interventions (such as emergency gas-plant activation), while recall-prioritized *High*-category alerts ensure that the operationally most dangerous periods are never missed, directly lowering the risk of unplanned load shedding.

3. Scope, Constraints & Risks

3.1 In-Scope

- **Historical data:** Red Eléctrica provides public access to data for real, scheduled and programmed energy demand (MW), energy generation by source, energy storage, and CO₂ emissions (t CO₂ eq/MWh) per source with a 5 minute frequency.
- **REE demand forecast:** Used as an input feature capturing intraday scheduling deviations, contributing to the next-day risk prediction.
- **Grid Risk Index:** Continuous composite score derived from operational stress variables. *Low*, *Medium*, and *High* labels are applied to predicted outputs for operational interpretation.
- **Driver attribution** per *High* category, showing which signal dominates grid risk.
- **Weekly policy digest** for Ministry and CNMC stakeholders: recurring risk patterns, dominant causes, and infrastructure implications.

3.2 Out-of-Scope

Four related tasks are deliberately excluded. **Automatic dispatch** requires safety certification governed by REE's grid-code operating procedures, which fall outside this project's remit. **Price forecasting** requires OMIE wholesale market data under separate terms and represents a distinct problem with different stakeholders. **Sub-regional breakdown** requires network-topology data unavailable in the REE dataset. **Infrastructure investment decisions** remain with policy makers; this model provides evidence and risk signals, not capital-allocation directives.

3.3 Constraints

- **Data access:** The REE dataset is publicly available with no API key, no authentication, and no rate limits; it is fixed for the duration of this project to remove infrastructure complexity.
- **Computation:** The model must run locally; no cloud GPU instances or specialized hardware. LightGBM is selected precisely because it completes training and inference in seconds on a standard laptop CPU.
- **Timeline:** 14 days to a demonstrable working model that is honest about its scope.
- **Labeling:** Labels are programmatically derived via PCA (see Section 4.2); no human annotation is required.

3.4 Risks & Trade-offs

Table 1 summarizes the four key risks identified, their potential impact, and the mitigation strategy adopted.

Table 1. Identified risks, failure modes, and mitigation strategies.

Risk	What Could Go Wrong	Mitigation Strategy
False Alarm Rate	If the model flags <i>High</i> risk too often and those alerts prove incorrect, operators will stop paying attention, even real alerts get ignored.	<i>Medium</i> acts as a buffer. <i>High</i> triggers only above a set confidence threshold; borderline signals remain <i>Medium</i> until the pattern is clear.
Model Generalisation	Trained on historical data only, the model may learn period-specific patterns and perform poorly on future conditions, especially as Spain adds renewable capacity.	Strict temporal train/test split. Evaluate on the most recent data portion. Flag any feature whose importance shifts significantly as a signal for future retraining.
Data Leakage	Adding features that rely on future information is easy. A model trained on leaked data performs well on assessments but fails in practice.	Automated leakage checking before every training job. If any feature's timestamp post-dates its label, training stops, no exceptions.
Fixed Threshold Misreads Hour-of-Day Context	A fixed MW threshold for <i>High</i> risk misclassifies periods where a smaller absolute gap is proportionally severe.	Gap is scored as a percentile rank within its hour-of-day distribution: proportional severity, not absolute MW. Stays adjusted as grid capacity grows.
Label Bias	The GRI is derived from REE operational data, which may systematically understate stress during high-renewable periods due to reporting conventions or data smoothing at 5-minute aggregation.	Cross-reference GRI high-scoring periods against REE's published annual incident reports. Flag any systematic divergence between model-identified high-risk days and reported incidents as a label-quality issue before final evaluation.
Scope Creep / Unrealistic Expectations	The 14-day timeline creates pressure to add features or outputs beyond the defined scope, risking an incomplete core deliverable.	Scope is frozen after Day 3. Any additions are logged as post-submission extensions and do not alter the evaluation targets defined in Section 2.

3.5 Anticipated Trade-offs

Two trade-offs have been explicitly evaluated in the design of this system. First, **recall vs. precision for the *High* class**: failing to flag a genuinely high-risk period exposes the grid to unplanned emergency interventions, whereas a false alarm causes operators to mobilize reserve capacity unnecessarily. Because the cost of a missed event exceeds the cost of a false alarm in this operational context, recall is prioritized over precision for the *High* category (see success criteria in Section 2). Second, **interpretability vs. accuracy**: more powerful but opaque ensemble methods (e.g. stacked deep learning models) were considered and rejected because REE operators require auditable, feature-level explanations for each prediction before acting on it. LightGBM is selected in part because it provides this transparency at a modest and acceptable cost to predictive accuracy.

4. Data Plan

This project relies exclusively on publicly available data from REE, ensuring **transparency, reproducibility, and institutional reliability**. The dataset includes near-real-time demand, day-ahead forecasts, and generation by source (wind, solar, hydro, combined-cycle gas turbines, nuclear) at five-minute resolution. The goal is not merely to compile raw variables but to transform them into

interpretable indicators of systemic stress, reflecting both the grid's exposure to imbalance and its capacity to respond.

4.1 Feature Engineering

Three core features operationalize the concept of systemic stress:

1. **Flexibility Share:** The proportion of total generation from dispatchable and adaptable sources (combined-cycle gas turbines and hydroelectric plants). Defined as $\text{FlexShare}_t = G_t^{\text{flex}} / G_t^{\text{total}}$. A lower value signals reduced adaptive capacity.
2. **Demand Forecast Error:** Quantifies deviation between expected and actual demand, computed as $\varepsilon_t = D_t^{\text{actual}} - D_t^{\text{forecast}}$. When realized demand exceeds forecasts, additional generation must be activated in real time, increasing system strain.
3. **Net Load** captures structural pressure on dispatchable generation, defined as $\text{NetLoad}_t = D_t^{\text{actual}} - (G_t^{\text{wind}} + G_t^{\text{solar}})$. Rapid changes (*ramps*) in net load reflect sudden increases in the burden placed on flexible resources.

4.2 Data Quality & Integrity

Although REE data are institutionally robust, missing values or reporting inconsistencies may occur during atypical conditions. The pipeline implements: **(i) temporal alignment** and consistency checks; **(ii) percentile-based proxy labels** for extreme operational states, avoiding direct reliance on rare blackout events; **(iii) class-imbalance** monitoring; and **(iv) strict temporal integrity** with chronological data splits to prevent leakage. Regarding **representativeness**, Spain's generation mix has shifted significantly over the dataset's time span; the model is therefore trained on the most recent data portion only, with performance evaluated on the latest held-out window. Regarding **systematic bias**, 5-minute aggregates smooth sub-minute frequency deviations that can precede stress events, a known resolution limitation. **Labeling** is entirely programmatic: the GRI is derived via PCA (Section 5.1) with no human annotation. The key challenge is that label quality depends on the input variable selection; to surface this, PCA weights are reported alongside the model and GRI values above the *High* threshold are cross-referenced against REE's published annual incident reports. Divergences between model-identified high-risk days and reported incidents are documented as labeling uncertainty rather than treated as model error.

5. Modelling Approach

The modeling strategy follows a two-stage approach: first building the composite Grid Risk Index, then training a short-term forecasting model on that index.

5.1 Index Construction

The Grid Risk Index is derived from the three main variables described in Section 4. To avoid arbitrary weight assignment, **Principal Component Analysis (PCA)** is applied to the standardized variables. The eigenvector associated with the largest eigenvalue (PC1) provides interpretable data-driven weights. These weights are normalized and used to produce a composite risk score $r_t \in [0, 1]$. PCA is estimated on training data only to prevent forward-looking bias.

5.2 Forecasting Framework

The forecasting task is formulated as a supervised learning problem: predict $r_{t+1\text{day}}$ using only information available up to time t . Three candidate models are evaluated in sequence before a final selection is made.

1. **Ridge Regression** serves as the primary interpretable baseline: linear, fast, and fully auditable. It establishes whether the relationship between lagged stress indicators and next-day risk is

sufficiently linear to require no further complexity.

2. **Random Forest Regressor** tests whether tree-based ensembles improve on the linear baseline without the sequential boosting overhead. It also provides a second source of feature-importance estimates for cross-validation.
3. **Gradient Boosted Trees (GBT) via LightGBM** (Ke et al., 2017) is the primary candidate, selected if it consistently outperforms the two simpler candidates in the held-out chronological test set by the margin defined in Section 2. **LightGBM captures nonlinear interactions** without strong parametric assumptions, trains sequentially so that each tree corrects prior residuals, and provides split-based feature-importance scores natively, allowing operators to identify which variable drove each *high-risk* prediction. **Interpretability** is required at this stage because REE operators must understand the dominant driver of a risk alert before committing to a dispatch decision.

A **persistence baseline** (next-day risk = current-day value) is used as the minimum usefulness threshold. The GBT model is considered useful only if it achieves an RMSE at least 15% lower than the persistence baseline on the chronological test set *and* recall for the *High* class exceeds 70%. A model failing either condition is not considered production-ready regardless of its absolute RMSE.

5.3 Model Evaluation

Performance is assessed on two levels:

- **Continuous output:** Evaluated with RMSE and MAE on a strictly held-out chronological test set.
- **Categorical threshold layer:** Applied post-hoc to the continuous output; evaluated using precision and recall for the *High* class. **Recall is the primary metric:** failing to detect a genuine *High-risk* period is the most operationally dangerous outcome.

6. System Preview & Project Plan

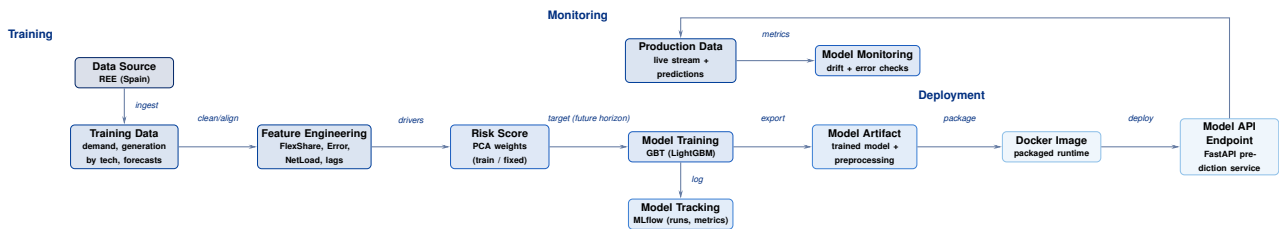


Figure 1. End-to-end MLOps pipeline for short-term grid risk forecasting.

6.1 Timeline

This is a 14-day proof of concept. The scope is intentionally honest: the model will run, produce real predictions, and be fully demonstrable. Table 2 details deliverables and done-criteria per phase.

Table 2. 14-day project timeline with deliverables and completion criteria.

Days	What Gets Done	Done When...
1 – 3	Download REE dataset, load all feeds, and merge into a single clean table. Compute the Grid Risk Index for the first time; check label balance and adjust thresholds if needed.	Clean merged dataset produced. Confirmed label balance.
4 – 6	Feature-engineering pipeline: lag features, velocity, cyclical encoding. Leakage unit test written and passing. Baselines trained and benchmarked.	Baseline scores documented. Leakage checks pass. No open blockers entering week two.
7 – 9	GBT trained on time-based split. Threshold adjustment for <i>High</i> precision/recall balance. RMSE computed on continuous score.	Model beats baseline. Risk Analysis and Classification produced.
10 – 13	Streamlit dashboard built on held-out predictions. Policy digest assembled from real model output. Full demo rehearsed as a team before day 13 ends.	Dashboard works. Someone outside the team understands it without instruction. Report complete. Demo runs cleanly.
14	Final submission.	All artefacts submitted.

6.2 Team Roles

- **Nicolás & Salah:** Data preparation, feature store, and leakage checks; verify data cleanliness before training begins.
- **Alberto & Gilles:** Modelling: GBT training, threshold adjustment, and evaluation metrics.
- **Madelyn:** Streamlit application and policy digest template.
- **All members:** Report writing, with cross-review of all figures and results before finalisation.

References

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